Optimization

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Outline

Motivation

Entropy

Conditional Entropy and Mutual Information

Cross-Entropy and KL-Divergence

Let’s work on this subject in our Optimization lecture

Cross Entropy

Cross Entropy: The expected number of bits when a wrong distribution Q is assumed while the data actually follows a distribution P

actual pdf

predicted

objective function

loss function

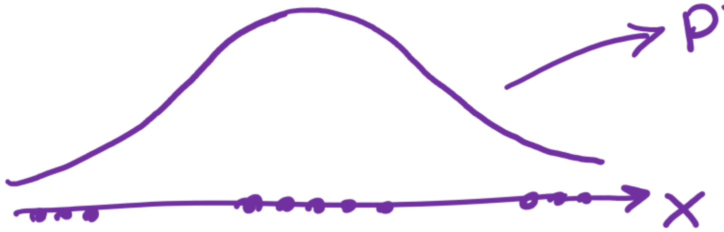
$$\boxed{H(p,q) =} - \sum_{x \in \mathcal{X}} p(x) \log q(x) = \underline{H(P)} + \underline{KL[P][Q]}$$

This is because:

$$H(p,q) = \mathrm{E}_p[l_i] = \mathrm{E}_p\left[\log \frac{1}{q(x_i)}\right]$$

$$H(p,q) = \sum_{x_i} p(x_i) \log \frac{1}{q(x_i)}$$

$$H(p,q) = -\sum_{x} p(x) \log q(x).$$



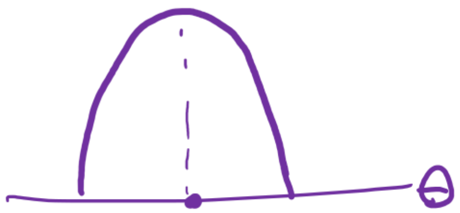
predicted pdf $(Q) = f(x^{\{i\}}|\theta)$ -

Objective function

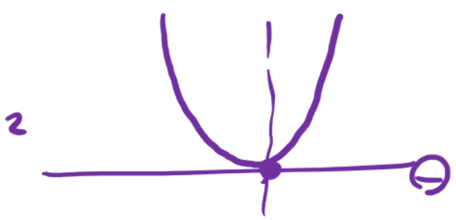
$$L(\theta|x) = \prod_{i=1}^{N} f(x^{\{i\}}|\theta)$$

$$\boxed{\text{Max} \quad \ell(\theta|x) = \sum_{i=1}^{N} \log f(x^{\{i\}}|\theta)}$$

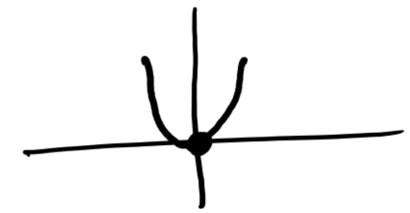
$f(x) = -x^2$



$f(x) = x^2$



$f(x) = \frac{x^2}{2}$



Min negative log likelihood

$$\text{Min} \quad \underbrace{\ell(\theta|x)}_{\in \mathbb{R}} \doteq -\sum_{i=1}^{N} \log f(x^{\{i\}}|\theta)$$

$\frac{1}{N}\sum(\cdot) = E[\cdot]$ $\quad g(x)$ $\quad g(x)$

$$\text{Min} \quad \ell(\theta|x) = \frac{1}{N}\sum\left(-\log f(x^{\{i\}}|\theta)\right) = E\left[-\log \underline{f(x^{\{i\}}|\theta)}\right]$$

avg negative log likelihood $= CE$

$$= E[-\log Q]$$

$$E[-\log P + \log P - \log Q]$$

$$\log P - \log Q = \log \frac{P}{Q}$$

$$= -\log \frac{Q}{P}$$

$$E[A+B+C] = E[A] + E[B+C]$$

$$E[g(x)] = \sum p(x)\, g(x)$$

$$\underbrace{E[-\log P]}_{} + E\left[-\log \frac{Q}{P}\right]$$

$$\downarrow$$

$$\sum P(x) * (-\log P(x)) + \sum P(x) * \left(-\log \frac{Q(x)}{P(x)}\right)$$

$$\sum P(x) \log_2 \frac{1}{P(x)} - \sum P(x) \log \frac{Q(x)}{P(x)}$$

$$H(P) + KL[P][Q] = CE = \frac{1}{N} \sum -\log f(x^{\{i\}} | \theta)$$

$$\cancel{-\sum P(x) \log P(x)} - \sum P(x) \log Q(x) + \cancel{\sum P(x) \log P(x)}$$

$$-\sum P(x) \log Q(x) = CE$$

Labeling target values
Label encoding (ordinal) and One-hot encoding

$$X = \begin{bmatrix} \text{height} & \text{weight} & \text{age} & \cdots \\ & & & \\ & & & \end{bmatrix}_{n\times d} \quad y_a = \begin{bmatrix} \text{fish} \\ \text{cat} \\ \text{dog} \\ \text{fish} \\ \vdots \end{bmatrix}_{n\times 1} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \\ \vdots \end{bmatrix} \rightsquigarrow \text{ML} \rightsquigarrow \hat{y}_p = \begin{bmatrix} 0.5 \\ 1.2 \\ 1.8 \\ \vdots \end{bmatrix}$$

$$\text{fish} = [1 \quad 0 \quad 0] \qquad \text{cat} = [0 \quad 1 \quad 0] \qquad \text{dog} = [0 \quad 0 \quad 1]$$

$$y_a = \begin{bmatrix} [1 & 0 & 0] \\ [0 & 1 & 0] \\ [0 & 0 & 1] \\ [1 & 0 & 0] \\ & \vdots & \end{bmatrix} \rightsquigarrow \text{ML} \rightsquigarrow \hat{y}_p = \begin{bmatrix} [78 & 20 & 10] \\ [2 & 92 & 3] \\ [2 & 3 & 60] \\ & \vdots & \end{bmatrix} \rightsquigarrow \text{Softmax} \rightsquigarrow \begin{array}{ccc} 0 & 1 & 0 \\ 1 & 0 & 0 \\ \end{array} \begin{bmatrix} [0.8 & 0.1 & 0.1] \\ [0.2 & 0.7 & 0.1] \\ [0.3 & 0.1 & 0.6] \\ & \vdots & \end{bmatrix}$$

$$\text{Max} \sum_{i=1}^{N} y_a^{\{i\}} \cdot \hat{y}_p^{T\{i\}} = \underbrace{[1 \quad 0 \quad 0] \begin{bmatrix} 0.8 \\ 0.1 \\ 0.1 \end{bmatrix}}_{x^{\{i\}}} \begin{matrix} 1 \\ 0 \\ 0 \end{matrix} + \cdots\cdots + \underbrace{\cdots}_{x^{\{n\}}} \Rightarrow \text{maximum possible is} = N$$

$$\text{Min} \sum_{i=1}^{N} \left\| y_a^{\{i\}} - \hat{y}_p^{\{i\}} \right\|_2^2 = \Big[\overbrace{(1-0.8)^2 + (0-0.1)^2 + (0-0.1)^2}^{x^{\{i\}}} + \cdots\cdots + \overbrace{\cdots}^{x^{\{n\}}} \Big]$$

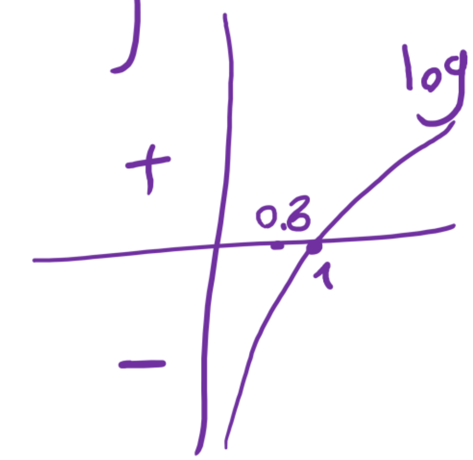
Why Cross entropy and not simply use dot product?

$\mathbb{R}$ $\in$

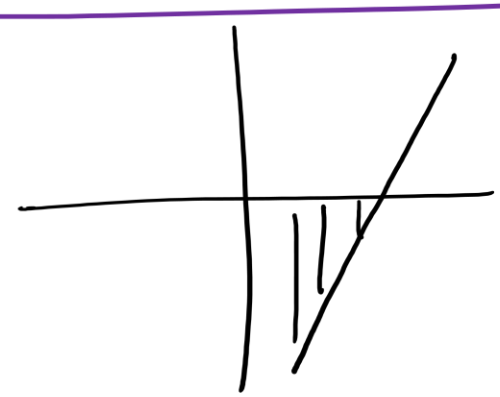
$$CE = -\sum_{i=1}^{N} y_a^{\{i\}} \log \hat{y}_p^{\{i\}}$$

$$y_a = \begin{bmatrix} [1 & 0 & 0] \\ \vdots & & \\ \vdots & & \end{bmatrix} \qquad \hat{y}_p = \begin{bmatrix} [0.8 & 0.1 & 0.1] \\ & \vdots & \\ & \vdots & \end{bmatrix}$$

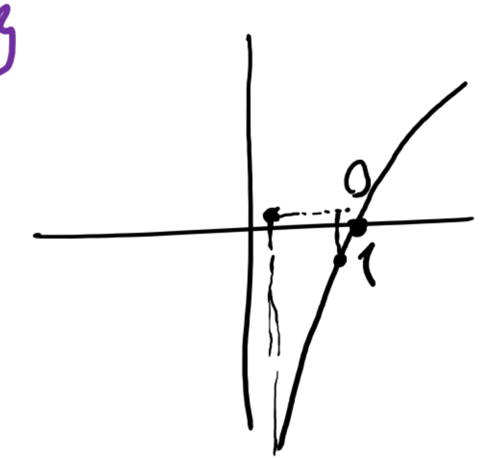
$$CE = -\left(\overset{y_a^{\{i\}}}{[1 \quad 0 \quad 0]} \underset{\hat{y}_p^{\{i\}}}{\begin{bmatrix} \log 0.8 \\ \log 0.1 \\ \log 0.1 \end{bmatrix}} + \cdots \cdots \right)$$



① $-\sum y_a^{\{i\}} \cdot \hat{y}_p^{\{i\}}$

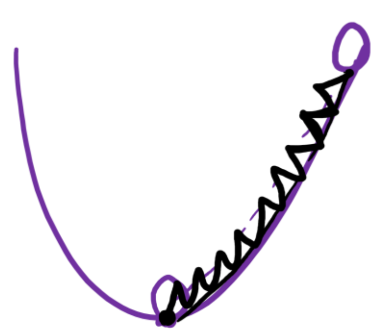


② $-\sum y_a^{\{i\}} \log \hat{y}_p^{\{i\}}$

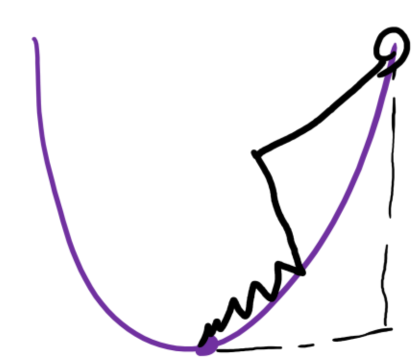


$f(x) = x^2$

$f'(x) = 2x = 0$



$f(x) = x^2$



Kullback-Leibler Divergence

Another useful information theoretic quantity measures the difference between two distributions.

$$\mathbf{KL}[P(S)\|Q(S)] = \sum_s P(s)\log\frac{P(s)}{Q(s)}$$

$$= \underbrace{\sum_s P(s)\log\frac{1}{Q(s)}}_{\text{cross entropy}} - \mathbf{H}[P] = H(P,Q) - H(P)$$

KL Divergence is a **KIND OF** distance measurement

Excess cost in bits paid by encoding according to Q instead of P.

$$-\mathbf{KL}[P\|Q] = \sum_s P(s)\log\frac{Q(s)}{P(s)}$$

log function is concave or convex?

$$\sum_s P(s)\log\frac{Q(s)}{P(s)} \le \log\sum_s P(s)\frac{Q(s)}{P(s)}$$

[By Jensen Inequality]

$$= \log\sum_s Q(s) = \log 1 = 0$$

So $\mathbf{KL}[P\|Q] \ge 0$. Equality iff $P = Q$

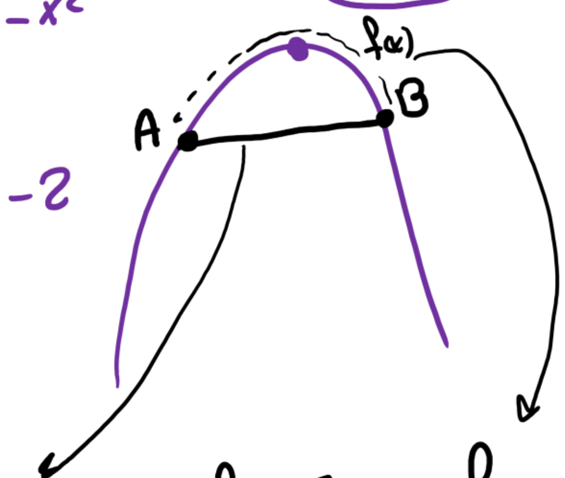
When $P = Q,\ KL[P||Q] = 0$

$f(x) = -x^2$

concave ~> strictly

$f''(x) = -2$

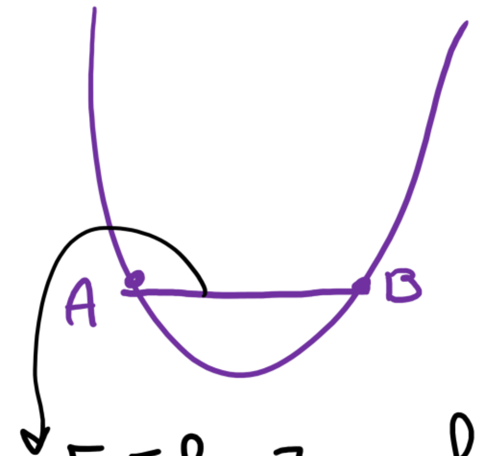
(1)



$E[f(x)] \leq f(E[x])$

$f(x) = x^2$

Convex



$E[f(x)] \geq f(E[x])$

$f(x) = x^2 \qquad X = [1, 2]$

$P(x) = [1/2, 1/2]$

$E[f(x)] = \sum p(x) f(x) =$

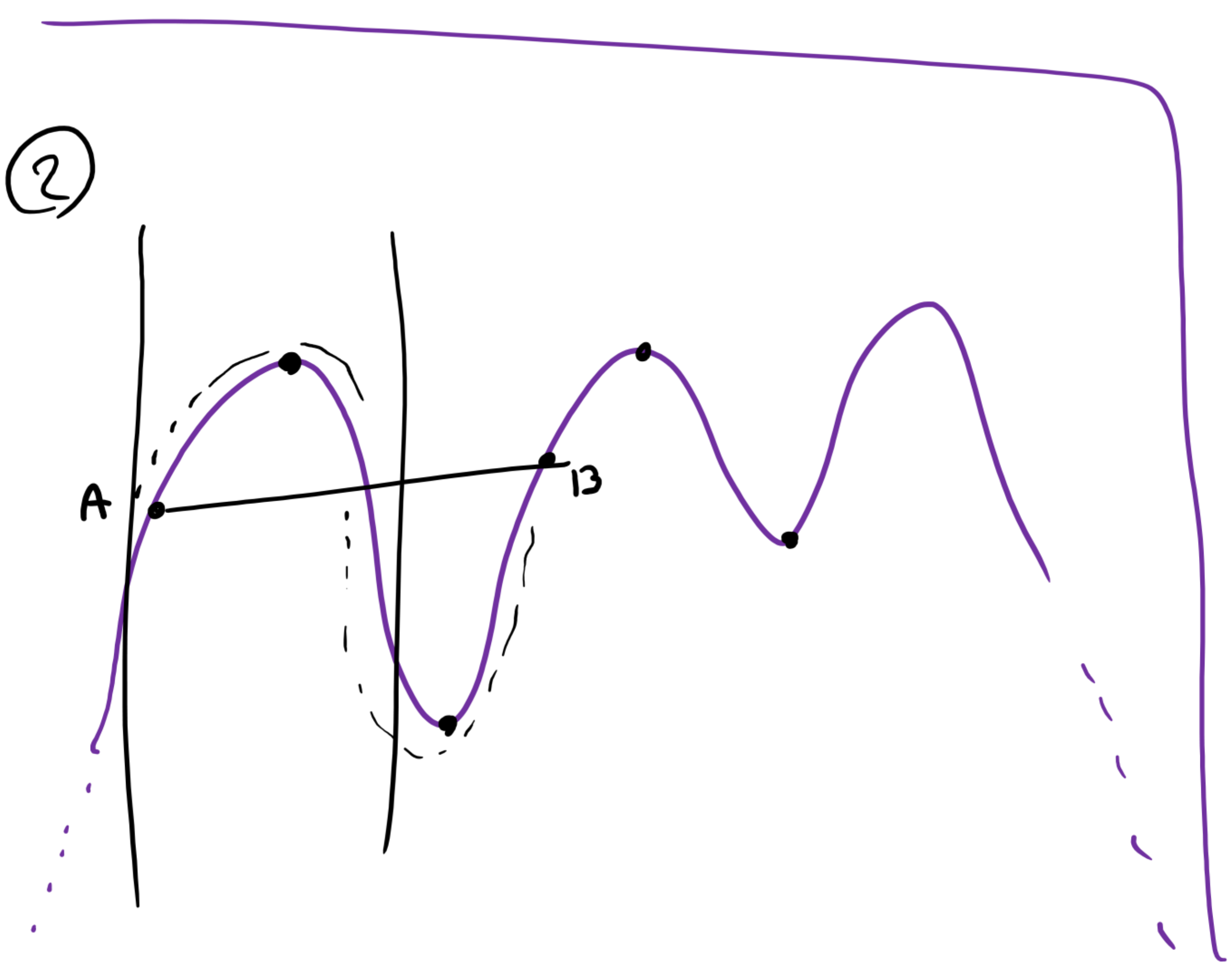
$P(x=1) f(x=1) + P(x=2) f(x=2)$

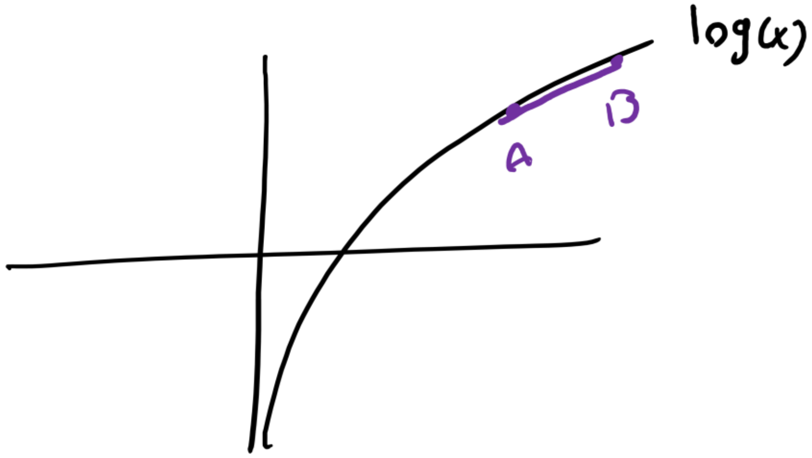
$\frac{1}{2} * 1 + \frac{1}{2} * 4 = 2.5$

$f(E[x]) = f(E[x] = 1.5) = 1.5^2$

$E[x] = \sum P(x) * X = \frac{1}{2} * 1 + \frac{1}{2} * 2 = 1.5$

(2)





$$E[f(x)] \leq f(E[x]) \Rightarrow E[\log x] \leq \log(E[x])$$

$$f(x) = \log x$$

$$-KL[P][Q] = \sum p(x) \log \overbrace{\frac{Q(x)}{P(x)}}^{g(x)} = \sum p(x) \log g(x)$$

$$E[\log g(x)] \leq \log E[g(x)]$$

$$\leq \log \sum p(x)\, g(x)$$

$$\leq \log \sum \cancel{p(x)} \frac{Q(x)}{\cancel{p(x)}}$$

$$\leq \log \sum Q(x)$$

$$-KL[P][Q] \leq \log 1 = 0 \Rightarrow KL[P][Q] \geq 0$$

objective function ← $f(x,y)$

S.t. $g(x,y)=0$ ~> Equality constraint

S.t. $h(x,y)\leq 0$ ~> Inequality constraint (KKT conditions)

$$f''(x,y) \rightsquigarrow H=\begin{bmatrix} \frac{\partial^2 f(x,y)}{\partial x^2} & \frac{\partial^2 f(x,y)}{\partial x \partial y} \\ \frac{\partial^2 f(x,y)}{\partial y \partial x} & \frac{\partial^2 (x,y)}{\partial y^2} \end{bmatrix}$$

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① Linear programming

$f(x,y) = x+y$

S.t. $x+y=0$

② Quadratic 〃

$f(x,y)=x^2+y^2$

S.t. $x+y=0$

③ non-linear 〃

$f(x,y)=x^3+y^2$

S.t. $x^2+y=0$

$$f(M,S) = 6M^2 + 3S^2$$

$M =$ # hours you study ML per day

$S =$ " " you sleep " "

$$\frac{\partial f(M,S)}{\partial M} = 0 \Rightarrow 12M = 0 \Rightarrow M = 0$$

$$\frac{\partial f(M,S)}{\partial S} = 0 \Rightarrow 0 + 6S = 0 \Rightarrow S = 0$$

$f(M,S) = 6M^2 + 3S^2$ → Objective function

$\delta, \alpha, \beta \rightsquigarrow$ Lagrange multiplier

Constraint function

s.t. $M + S = 24$

$g(M,S) = M + S - 24$

$$L(M,S,\delta) = f(M,S) - \delta\, g(M,S) = 6M^2 + 3S^2 - \delta(M+S-24)$$